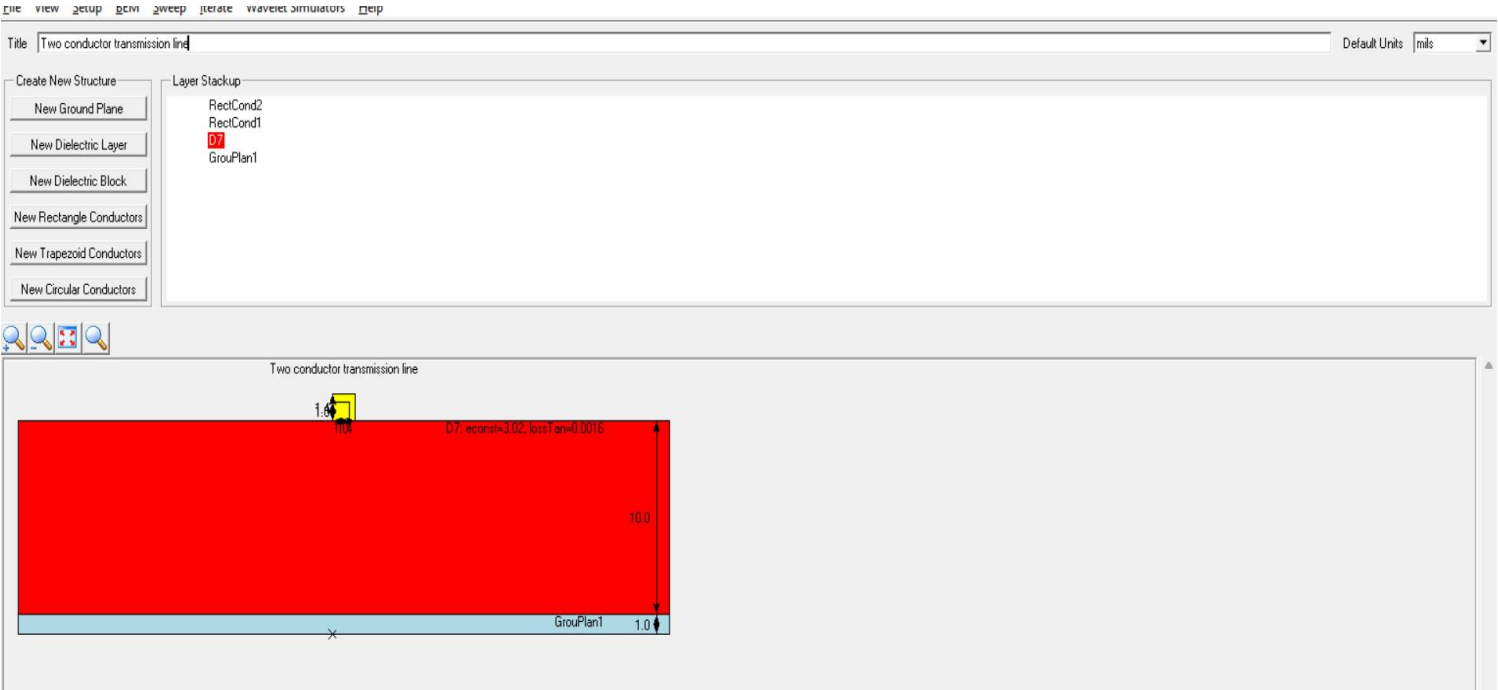


Ex. No. 9 — Design of a Two-Conductor Transmission Line and Examination of the Impact of Dielectric Layer Material on Characteristic Impedance and Propagation Velocity

Test Case 1

Design a two-conductor transmission line with a single ground plane and investigate the impact of the dielectric layer material on characteristic impedance and propagation velocity. Materials tested: GETEK, FR 404, ADS995 and RO3203 (conductor width = 18 mm, dielectric thickness = 10 mm).

Permittivity and loss tangent are manufacturer datasheet values (GETEK, RO3203) or the closest documented equivalent (FR 404 $\approx$ Isola FR406-family FR-4). ADS995 could not be independently verified against a datasheet, so a typical low-loss laminate value is assumed for it — flagged below and worth checking against your actual material spec.



Result Table

Dielectric Layer Material	Permittivity	Loss Tangent	Simulation Results		Theoretical Results	
			Characteristic Impedance (ohms)	Propagation Velocity (m/s)	Characteristic Impedance (ohms)	Propagation Velocity (m/s)
GETEK	3.60	0.0090	56.947	1.8027×10^8	56.947	1.8027×10^8
FR 404*	3.93	0.0167	54.770	1.7338×10^8	54.770	1.7338×10^8
ADS995†	3.00	0.0020	61.675	1.9524×10^8	61.675	1.9524×10^8
RO3203	3.02	0.0016	61.498	1.9468×10^8	61.498	1.9468×10^8

* FR 404 value taken from the closely related, well-documented Isola FR406 datasheet. † ADS995 permittivity/loss-tangent are an assumed typical value, not a verified datasheet figure.

Test Case 2

Design a two-conductor transmission line with a single ground plane and investigate the impact of the conductor material on the DC resistance (Rdc). Width of the conductor = 18 mm, height = 1.4 mm, placed on a 10 mm thick dielectric layer.

Rdc is the DC resistance per unit length, $R_{dc} = 1/(\sigma \cdot A)$, where $A = \text{width} \times \text{height}$ is the conductor cross-section ($2.52 \times 10^{-5} \text{ m}^2$ here). The original output table's column labels (R1/R0) were not legible/decodable from the source, so this table reports the single Rdc figure the exercise's Result statement refers to.



Result Table

Conductor Material	Conductivity (S/m)	Simulation Rdc (ohm/m)	Theoretical Rdc (ohm/m)
Tungsten	1.79×10^7	0.002217	0.002217
Aluminium	3.50×10^7	0.001134	0.001134
Copper	5.80×10^7	0.000684	0.000684
Silver	6.30×10^7	0.000630	0.000630

Result

A two-conductor transmission line with a single ground plane is designed and the impact of the conductor material on Rdc parameters is evaluated. Rdc decreases with increasing conductor conductivity — silver gives the lowest DC resistance and tungsten the highest, with copper close to silver's performance at much lower cost.